Young's Double Slit Experiment:

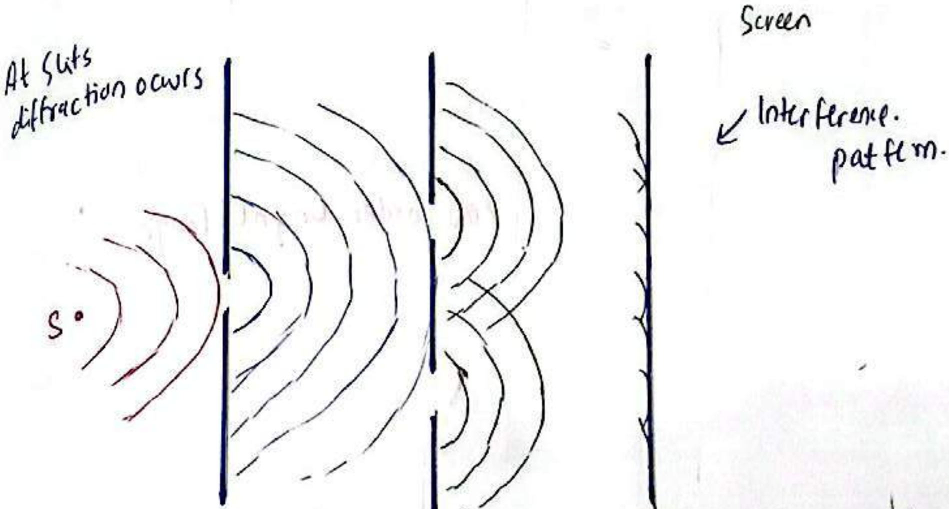
- 1801
- Prove the wave nature of light
- idea use (Huygens).

Principle: Division of wave fronts or wave lengths.

- young used one source of light because According to young's two different source of light cannot be coherent.

After discovery of laser we can say that,

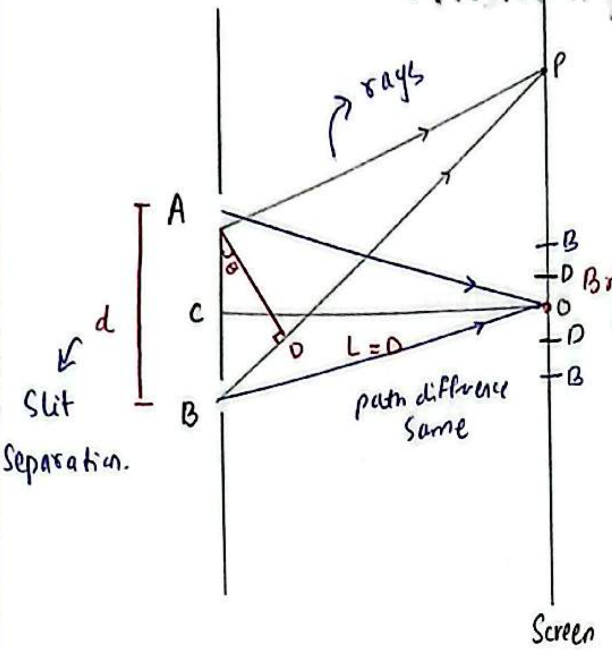
- two different source of light cannot be coherent except laser. (In era of young's laser did not exist)

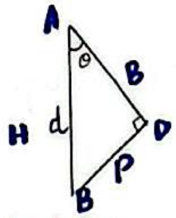


- In young double slit Experiment the process taking place was Interference & diffraction (both)
- In young double slit Experiment the process under observation was Interference.
- young double slit experiment is study of or experiment of Interference

Young's Double Slit Experiment:

$BD = \Delta S$ = path difference $= d\sin\theta$.





$$\sin\theta = \frac{P}{H}$$

$$\sin\theta = \frac{BD}{AB}$$

$$\sin\theta = \frac{BD}{d}$$

$$BD = d\sin\theta$$

$$\boxed{\Delta S = d\sin\theta}$$

For constructive interference :-

because of constructive interference

$d\sin\theta = \underbrace{m}_{\text{order of bright fringe (interference line)}}\lambda$

$m = 0, \pm1, \pm2, \ldots\ldots$

$d\sin\theta = 0, \lambda, 2\lambda, 3\lambda, \ldots$

At centre, Bright fringe will made and we call it zero order bright fringe.

For destructive interference :-

$d\sin\theta = (\underbrace{m}_{\text{order of dark fringe}} + \frac{1}{2})\lambda$

$d\sin\theta = (2m+1)\lambda/2$

$m = 0, \pm1, \pm2, \ldots\ldots$

$d\sin\theta = \lambda/2, 3\lambda/2, 5\lambda/2, \ldots$

odd multiple of $\lambda/2$.

order :-

1st bright fringe $\boxed{m=1}$

2nd bright fringe $\boxed{m=2}$

centre bright fringe $\boxed{m=0}$

1st order dark
$m=0$, $d\sin\theta = \lambda/2$

2nd order dark
$m=1$

Zero order dark fringe does not exist but $m=0$ do.

MCQ # 01 7^{th} order dark fringe....

$m = 6$

$d\sin\theta = 13\lambda/2$

mcq#(07) if no of fringes are equal to 10 above the 3rd order bright fringe, the total no of fringes are?

(a) 10 (b) 20 (c) $\boxed{21}$ (d) 11

Note: Fringes above origin will be equal to fringes below origin.

total no. of fringes = 10+10+1 = 21

⟹ Bright fringes > dark fringes.

central point (YDSE)

bright (replacement word for any particular color)

↓

monochromatic light

e.g if we have used green light (monochromatic) then central spot will also be green.

↓

if we use monochromatic light then fringes will be bright & dark.

if there are two sources of light they must be monochromatic if there is only one source of light that could be monochromatic or polychromatic.

white

$\xrightarrow[\lambda\uparrow, n\downarrow]{VIBGYOR}$

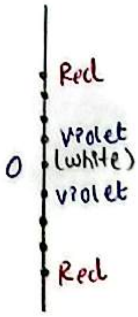
↓

white light (mixture of 7 colors)

central spot will be white.

↓

if we use white light then coloured fringes will be produce.



white light interference occur when same color of light will meet.

V-V | G-G | Y-Y

V-G → no interference

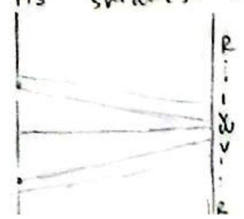
interference will occur but whether it's constructive or destructive depends on path difference.

$\uparrow n \propto \frac{1}{\lambda\downarrow}$

Refractive index

parameter which decides the bending of light

So violet light will bend greatest because of its smallest wavelength among all.



fringes.

int (YDSE)

if there are two sources of light they must be monochromatic if there is only one source of light that could be monochromatic or polychromatic.

white

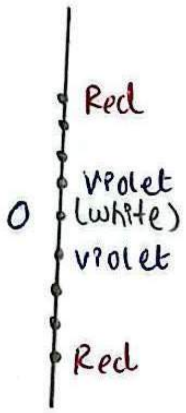
VIBGYOR

$\lambda\uparrow, n\downarrow$

white light (mixture of 7 colors)

central spot will be white.

if we use white light then coloured fringes will be produce.



White light interference occur when same color of light will meet.

V-V | G-G | Y-Y

V-G → no interference

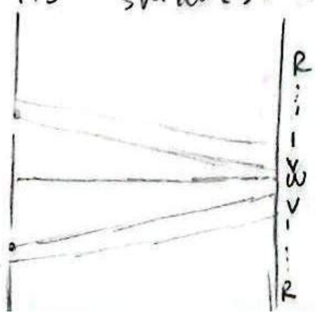
interference will occur but whether its constructive or destructive depends on path difference.

$\uparrow n \propto \frac{1}{\lambda\downarrow}$

Refractive index

perimeter which decides the bending of light

So violet light will bend greatest because of its smallest wavelength among all.



MCQ# 03 In YDSE tenth fringe of 400nm $\boxed{\text{concide}}$ with 8th fringe of wave length "λ" find λ = ?

Key word

$d\sin\theta = \Delta S$

$m_1 = 10$

$\lambda_1 = 400nm$

$m_2 = 8$

$\lambda_2 = ?$

For constructive interference :-

~~when~~ $d\sin\theta = m\lambda$

when waves concides their path difference becomes constant.

constant = $m\lambda$

$$\boxed{m \propto \frac{1}{\lambda}}$$

$$\frac{m_1}{m_2} = \frac{\lambda_2}{\lambda_1}, \quad \frac{m_1\lambda_1}{m_2} = \lambda_2, \quad \frac{10(400)}{8} = \boxed{500\,nm}$$

MCQ # 04 Find the angular position of 3rd order maxima in interference pattern of YDSE.

$\theta = ?$ Bright fringe

$\boxed{a}$ $\sin^{-1}(3\lambda/d)$

(b) $\sin^{-1}(5\lambda/d)$

(c) $\cos^{-1}(3\lambda/d)$

(d) $\cos^{-1}(\lambda/d)$

$m = 3$

$d\sin\theta = m\lambda$

$\sin\theta = \frac{m\lambda}{d}$

$\sin\theta = 3\lambda/d$

$\theta = \sin^{-1}(3\lambda/d)$

if third order minima :-

$m = 2$

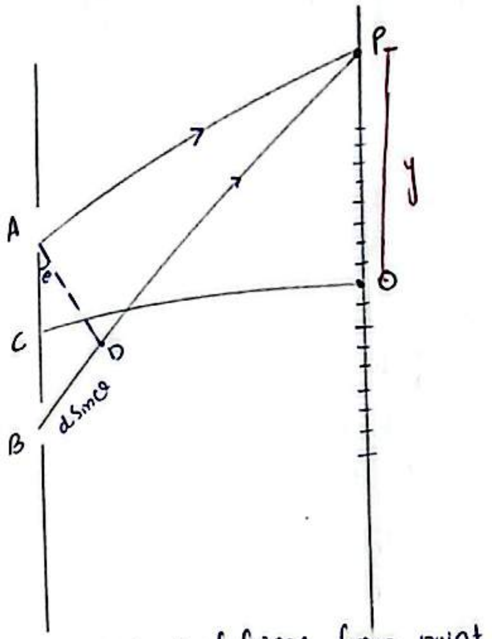
$d\sin\theta = \left(m + \frac{1}{2}\right)\lambda$

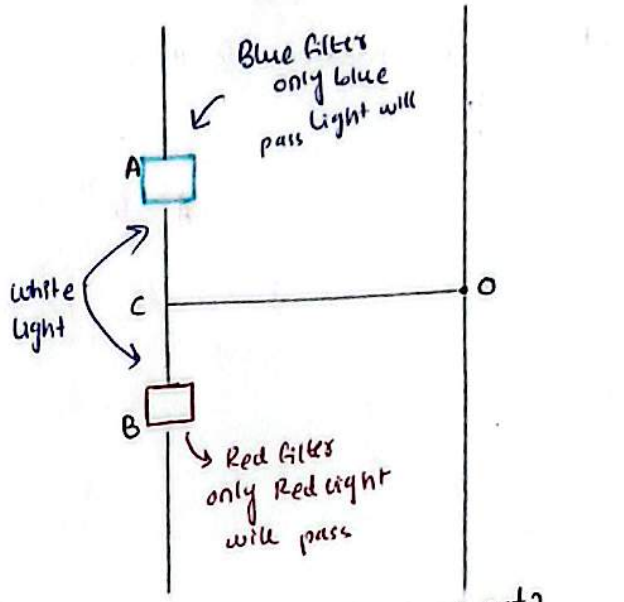
$d\sin\theta = (2m+1)\lambda/2$

$d\sin\theta = 5\lambda/2$

$\sin\theta = 5\lambda/2d$

$\theta = \sin^{-1}(5\lambda/2d)$





will interference occur or not?

↳ No interference will occur when two different colors combine on screen. (B-R)

distance of fringe from point O.

Position of Fringes:

Bright fringe

if at point "P" constructive interference has occured.

$$y_B = \frac{m\lambda L}{d}$$

dark fringe

if at point "P" dark fringe has ocurred.

$$y_d = (m+\tfrac{1}{2})\frac{\lambda L}{d}$$

MCQ # 05 calculate position of 2nd order ~~dark~~ bright fringe.

$$y_2 = \frac{(2)\lambda L}{d}$$

$$y_{2(B)} = \frac{2\lambda L}{d}$$

MCQ # 06 calculate position of 2nd order dark fringe.

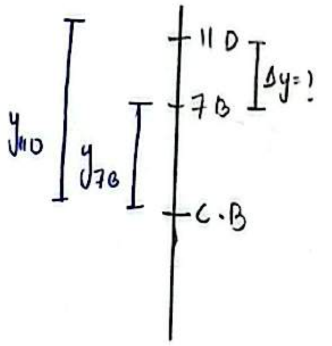
$$y_d = (1+\tfrac{1}{2})\frac{\lambda L}{d}$$

$$y_d = \frac{3\lambda L}{2d}$$

MCQ # 07 if $d = 1\,mm$, $L = 0.5\,m$, $\lambda = 5000\,A^\circ$. Find the Separation between 7^{th} maxima & 11^{th} minima of the same side of central maxima.

$$A^\circ = 10^{-10}$$

$$\lambda = 5000\,A^\circ = 5000 \times 10^{-10}$$



$$\Delta y = y_{11\,dark} - y_{7\,Bright}$$

$$y_{7B} = \frac{7\lambda L}{d}$$

$$y_{11D} = \frac{21\lambda L}{2d}$$

$$\Delta y = \frac{21\lambda L}{2d} - \frac{7\lambda L}{d}$$

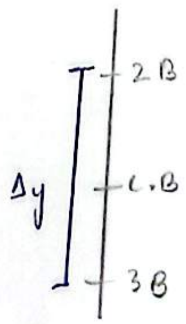
$$\Delta y = \frac{21\lambda L - 14\lambda L}{2d} = \frac{7\lambda L}{2d}$$

$$\Delta y = \frac{7(5000 \times 10^{-10})(5)}{(10)(2)(1 \times 10^{-3})}$$

$$\Delta y = 8.75 \times 10^{-3}\,m$$

$$\Delta y = 8.75\,mm.$$

MCQ # 08 Find the separation between 2nd maxima & 3rd maxima on opposite sides of central maxima.



$$\Delta y = y_{2B} + y_{3B}$$

$$\Delta y = \frac{2\lambda L}{d} + \frac{3\lambda L}{d}$$

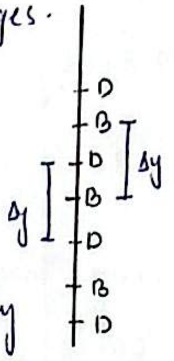
$$\Delta y = \frac{5\lambda L}{d}$$

$\Delta y =$

Fringe Spacing:- Separation between two consecutive bright or two consecutive dark fringes.

$$\Delta y = \frac{\lambda L}{d}$$

$$\Delta y_{Bright} = \Delta y_{dark} = \frac{\lambda L}{d}$$



⇒ Interference fringes are equally spaced & equally bright & dark.

Separation between consecutive bright & dark fringe $= \frac{\lambda L}{2d}$

$\uparrow \Delta y \propto \lambda \uparrow$

$\lambda \propto 1/f$

$\Delta y \propto 1/f$

V I B G Y O R Δy_{max}

$\lambda\uparrow, f\downarrow$

Δy_{min}

Fringe Spacing will be maximum for Red light & minimum for violet light.

$\Delta y \propto L$ (Distance between slits & Screen)

$\Delta y \propto 1/d$ (slit Separation)

which parameter should be decreased so fringe Spacing (Δy) be increased.

d (slit separation)   f (frequency)

MEDIUM CHANGE:-

$$\Delta y = \frac{\lambda L}{nd}$$

refractive index.

$\Delta y \propto 1/n$

↳ if we perform YDSE in water instead of Air- what is effect on Δy?

$n_{water} > n_{air} > n_{vaccum}$

$\downarrow \Delta y \propto 1/n\uparrow$

Fringe Spacing will decrease.

↳ if we perform young's double slit experiment in vaccum instead of Air what is effect on Δy?

$$\uparrow \Delta y \propto 1/n \downarrow$$

fringe spacing will increase.

Ratio of fringe spacing between two consecutive bright & two consecutive dark fringe $\frac{\Delta y_B}{\Delta y_D} = 1$ because $\Delta y_B = \Delta y_D = \frac{\lambda L}{d}$

MCQ # 08 In YDSE if $\lambda = 6000A°$ then 1st maxima is obtained at 0.2m from central maxima if $\lambda' = 4000A°$ where you will find the 4th minima.

ⓐ $\boxed{0.466m}$

ⓑ 412m

ⓒ 512m

ⓓ 0.512m

$$y_B = \frac{m\lambda L}{d}, m = 1$$

$$y_B = \frac{\lambda L}{d}$$

$$0.2 = \frac{(6000 \times 10^{-10}) L}{d} \quad \text{——— ①}$$

$$y_D = (m + 1/2)\frac{\lambda L}{d}$$

$$y_D = \frac{7\lambda L}{2d}$$

$$y_D = (m + 1/2)\lambda L/d$$

$$y_{4D} = \frac{7(4000 \times 10^{-10}) L}{2d}$$

dividing :-

$$\frac{y_{4D}}{0.2} = \frac{7(4000 \times 10^{-10})}{2(6000 \times 10^{-10})}$$

$$\frac{y_{4D}}{0.2} = \frac{7(4 \times 10^3 \times 10^{-10})}{6 \times 10^3 \times 10^{-10}} = \frac{7(4 \times 10^{-7})}{6 \times 10^{-7}} = \frac{7(\cancel{4} \times 10^{-7+7})}{2(\cancel{6}\,3)} = \frac{14}{2(3)}(0.2)$$

$$y_{4D} = \cancel{\ldots} \quad \frac{14}{6}(0.2) = 0.466$$

$$15 \overline{)140} = 0.93$$
$$135$$

Be CRAZY, BE STUPID, BE SILLY, BE WEIRD, BE WHATEVER.

INTERFERENCE:

∴ Two identical waves → propagate in same medium

→ (λ, f, T, V, medium) same

→ Amplitude may or may not be equal.

- Travelling in same direction. | if opposite direction then resulting wave will be stationary waves.

- Superpose

- Principle of superposition is applicable.

Application of principle of superposition:

Interference | Stationary waves | beats.

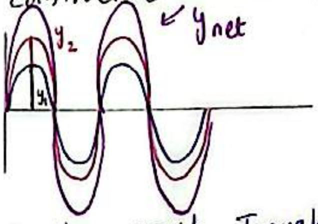
$\vec{y}_{net} = \vec{y}_1 + \vec{y}_2 \ldots$ (vector sum of individual waves)

amplitude of resultant wave.

Note: In Interference energy is conserved but redistributed.

TYPES OF Interference

Constructive Interference



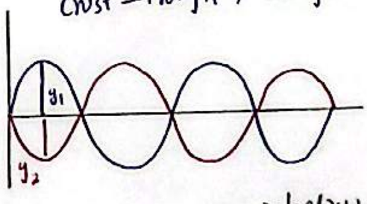
Crust – crust, Trough – Trough.

$y_{net} = y_1 + y_2$ → λ, f, T, V same as individual waves.

→ (Amplitude) ↑ (Resultant wave)

destructive interference.

Crust – Trough, Trough – crust.



$y_{net} = y_1 \ominus y_2$ → below mean position.

→ λ, f, V, T same as individual waves.

→ Amplitude of resultant wave

$\boxed{\text{Zero}}$ if two waves have same amplitude

$\boxed{\text{minimum}}$ if 2 waves have diff amplitude.

Silence zone in ocean are form due to destructive interference.

TYPES OF INTERFERENCE

Constructive Interference

path difference : (length (length measurement) in form of λ)

$\Delta s = m\lambda$ (± for origin position)

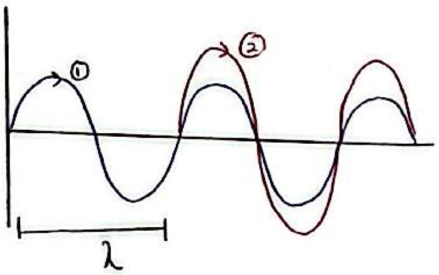
$m = 0, \pm 1, \pm 2, \pm 3, \ldots$

if waves meet here −1 | Origin | if waves meet here +1

$\Delta s = m\lambda$ (Integral multiple of λ)

$\Delta s = 0, \pm\lambda, \pm 2\lambda, \pm 3\lambda, \ldots$

Note : when we make diagrams ~~conditions~~ to understand conditions we always draw crust first.



path difference is of λ & constructive interference has occurred.

phase difference : (means angle)

$m = n$

$\varphi = 2m\pi$

$m = 0, \pm 1, \pm 2, \pm 3, \ldots$

$\varphi = 0, 2\pi, 4\pi, 6\pi, \ldots$

(even multiple of π)

Note : $\varphi = \pi/2$ rad (No interference)

in degrees

$\varphi = 0, 360^\circ, 720^\circ, \ldots$

Destructive Interference

path difference :

$\Delta s = (m + \frac{1}{2})\lambda$

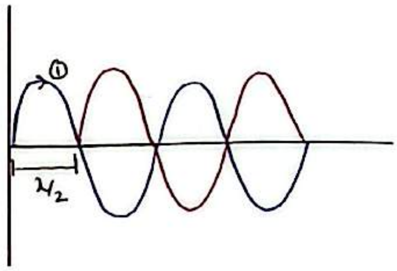
$\Delta s = (2m+1)\lambda/2$

$m = 0, \pm 1, \pm 2, \ldots$

$\Delta s = \pm \lambda/2, \pm 3\lambda/2, \pm 5\lambda/2, \ldots$

→ odd multiple of λ/2

Note : $\Delta s = \lambda/4$ (No interference)



path difference is of λ/2 & destructive interference has occurred.

phase difference :

$m = n$

$\varphi = (2m+1)\pi$

$m = 0, \pm 1, \pm 2, \pm 3, \ldots$

$\varphi = \pi, 3\pi, 5\pi, 7\pi, \ldots$

(odd multiple of π)

$\varphi = 180^\circ, 3(180), 5(180).$

phase difference $\phi = \frac{2\pi x}{\lambda}$ path difference

Relation between path & phace difference.

MCQ #09 two waves having path difference of $\lambda/4$, what will be the phase difference?

"No interference will occure." but the question is to convert $\lambda \longrightarrow \phi$.

$$\phi = \frac{2\pi(\lambda/4)}{\lambda}$$

$$\phi = \pi/2 \text{ rad}, \quad \phi = 90^\circ$$

MCQ #10 two waves having phase difference of $\pi/4$ rad. what will be the path difference?

"No interference will occur." - - - -

$$\phi = \frac{2\pi x}{\lambda}$$

$$\pi/4 = \frac{2\pi x}{\lambda}$$

$$\lambda \times \pi/4 \times \frac{1}{2\pi} = x$$

$$x = \lambda/8$$

$$\lambda = 2\pi \text{ rad}$$

$$\pi/4 = \frac{2\pi x}{\lambda}$$

$$\lambda \times \pi/4 \times \frac{1}{2\pi} = x$$

$$x = \lambda/8$$

$\lambda = 2\pi$ rad

$\lambda = 360°$

TYPES OF INTERFERENCE

Constructive Interference

$A_{max} = A_1 + A_2$ (Amplitude)

$\therefore I \propto A^2$ (Acc to classical physics) In modern physics [illegible] of ~~intensity~~ gets changed & I = no of particles.

$$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$\boxed{I_{max} = I_1 + I_2 + 2\sqrt{I_1 I_2}}$$

Destructive interference

$A_{min} = A_1 - A_2$ (Amplitude)

$I \propto A^2$

$$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$\boxed{I_{min} = I_1 + I_2 - 2\sqrt{I_1 I_2}}$$

MCQ# (11) Two waves which has same "f", Intensities are in the ratio 9 : 16. Find the ratio of I_{max} to I_{min} of resultant wave if these two waves superpose.

(a) 16 : 9 (b) 9 : 16 (c) 4 : 3 (d) $\boxed{49 : 1}$

$$I_1 = 9$$
$$I_2 = 16$$

$$I_{max} = (3+4)^2 = (7)^2 = 49$$
$$I_{min} = (3-4)^2 = (-1)^2 = 1$$

$$\frac{I_{max}}{I_{min}} = \frac{49}{1}$$

MCQ# (12) Two waves which have same "f", Amplitudes are in the ratio 3 : 5. Find the ratio of I_{max} to I_{min} of resultant wave if these two waves superpose.

(a) 1 : 16 (b) $\boxed{16 : 1}$ (c) 5 : 3 (d) 25 : 9.

$$A_{max} = 8$$
$$A_{min} = 2 \quad \text{(Large} - \text{Small)}$$

$$\frac{A_{max}}{A_{min}} = \frac{4}{1}$$

$$\because I \propto A^2$$
$$\frac{I_{max}}{I_{min}} = \frac{16}{1}$$

VImp Interference (principles)

Two

Division of amplitude
- Thin film
- Newtons Rings
- Michelson's Interferometer

Division of wave fronts / wavelengths.
- Young double slit experiment (YDSE)
- Fresnel's by prism
- Lloyd's mirrors.

Source for Interference:

Note: Both sound & light waves shows interference.

→ Same color of light

↳ monochromatic $\lambda = $ Same (OR) Single λ.

↳ coherent ($\Delta\phi = 0$, (OR) $\Delta\phi = $ constant) → even multiple of π (OR) odd multiple of π

$\Delta\phi = \pi/2$ rad & constant will they do interference? & are they coherent?

↳ no interference, not coherent

Secondary conditions:-

↳ There two source of light placed closed to each other so that their waves superpose with each other.

→ if not placed close to each other interference won't occur despite the fact that they are monochromatic & coherent.

↳ There should be slight difference in amplitude if difference is too much greater no interference will occur.

Above mentioned conditions are for 2 source of light.

one Source of light (single source) Always coherent.

↳ beneficial as compared to two sources.

↙ monochromatic

↘ polychromatic

interference on soap bubbles is due to white light coming from sun.

sun ⇒ single source

" ... sense of vision"

monochromatic

polychromatic

Always coherent.
→ benefitial as compared to two sources.

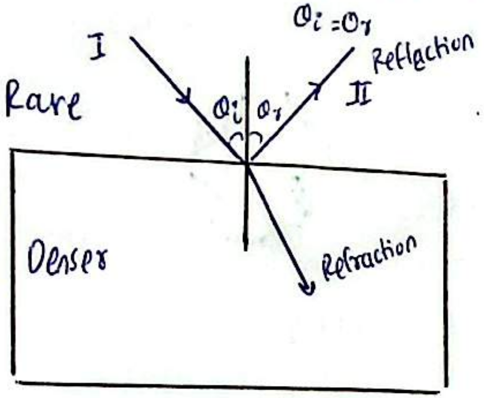
interference on Soap bubbles is due to white light coming from Sun
Sun ⇒ Single Source

LIGHT: "Form of energy which gives us sense of vision"

Nature, → wave nature (we study propagation of light)
→ Reflection, refraction, diffraction, polarization, Interference.

→ particle nature. (we study interaction of radiation with matter). Photoelectric effect, pair production., comptons effect.

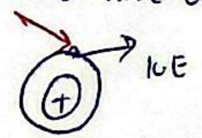
↓
Dual nature



Rare ⟶ Denser
• bend towards normal

Denser ⟶ Rare
• bend away from normal.

Glass atoms would have electrons



Light has given energy to that $\vec{e}$ & $\vec{e}$ has absorbed that energy.
& emit.
→ Studying matter particle behaviour.

MCQ # 13 The First Scientist who gave the idea of wave nature of light was.

(a) de-broglie
(b) Huygens
(c) Youngs
(d) Maxwell → Nature of light (theory) in 1873
↳ lights waves are electromagnetic waves.

MCQ # 14 The First Scientist who proved light to wave nature was.

(a) Einstens → light (particle nature) ↳ photons
(b) Newtons → corpuscles ↳ light (particle nature)
(c) Huygens
(d) Youngs.

MCQ # 15 Blue color of Sky is due to?

(a) Polarization
(b) Interference
(c) Diffraction

] wave nature of light — wave character $\propto \lambda$

(d) Scattering] particle nature of light — particle character $\propto 1/\lambda$

When light travels & collides with tiny particles, it changes direction — this is called Scattering.

Scattering $\propto 1/\lambda$ — Reason why blue & Red dominates.

V I $\boxed{B}$* G Y O $\boxed{R}$*

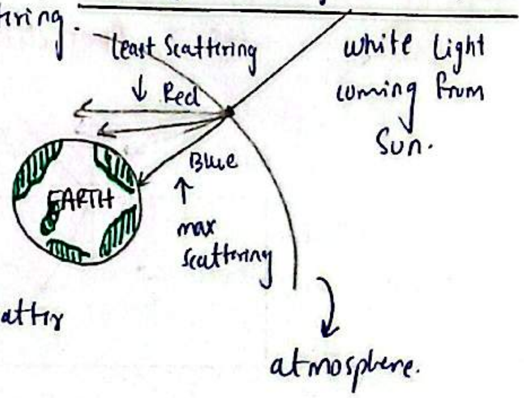
$\lambda\uparrow$, (WC)$\uparrow$, (PC)$\downarrow$

wave length pattern of visible light · 400nm — 700nm between these 300nm wavelength maximum portion of λ in shorter wavelength is of blue light & in longer wavelength it is by Red

→ violet color will have $\boxed{\text{maximum}}$ Scattering.

→ Red color will show $\boxed{\text{minimum}}$ Scattering.

Among Shorter wavelengths color Blue color dominates & Among longer wavelengths color Red color dominates.



Sunlight has many colors. Air molecules Scatter blue light more, MCQ # (16) if there would have been no atmosphere then the color of sky would seem to be?

(a) Blue (b) white (c) $\boxed{\text{Black}}$ (d) Red.

No Scattering will occur & no color will reach earth.

MCQ # (17) During sunset sky seems to be red, it is due to the reason at that time.

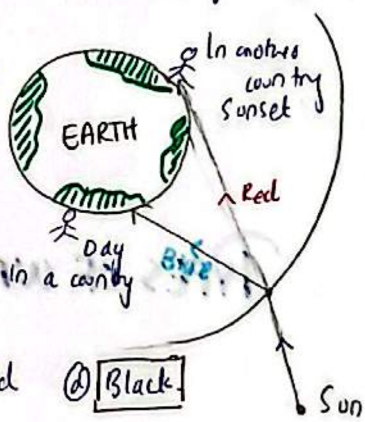
(a) Blue colour Scatter least
(b) Red colour Scatter most
] as λ fix these could not be true

(c) both (a) & (b)

$\boxed{\text{(d)}}$ Red Scatter least & blue most.

→ In some countries if there is day time, in other countries there would be evening time.

EARTH IS CONTINUOUSLY REVOLVING AROUND SUN.



MCQ # (18) A red flower kept in green light appears to be

(a) Red (b) Green (c) Greenish red (d) $\boxed{\text{Black}}$

we normally see things with the help of white light because white light comes from sun. white light consist of 7 colors & for example it has fallen on an object & it seems yellow to us it means that object has absorbed all the other colors except that yellow color & reflected that yellow color.

→ Red flower will appear red only when we see it through Red light or white light. Through all the other monochromatic lights, we will see it black.

↳ light of a single wavelength is called ⇒ monochromatic light

→ provide constant light due to nuclear Fussion reaction.

MCQ #(19) The twinkling of Stars is due to?

(a) constructive interference only.

(b) Destructive interference only.

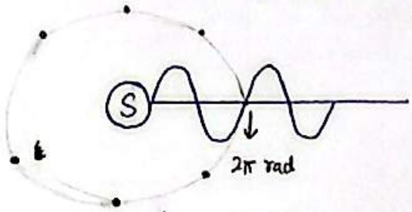
(c) Both constructive & destructive.

(d) $\boxed{\text{non uniform density of atomsphere.}}$

- when star light appears in front to us, atmosphere of it is very thin & its light passes through that thin atmosphere. & when we are unable to see that star, a thick atmosphere occur infront of it & its light don't pass through it

WAVE Fronts:

Light is emitting from Source of Light in all 3 dimensions.



All these black points have Same phase (angle) of 2π rad. After joining all these points we got a Surface.

Surface (All points on this Surface has Same phase).

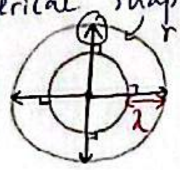
" An imaginary Surface on which all points have Same phase of vibration."

Note: wavefrons propagate with the → Speed of light.

TYPES OF WAVE Fronts:

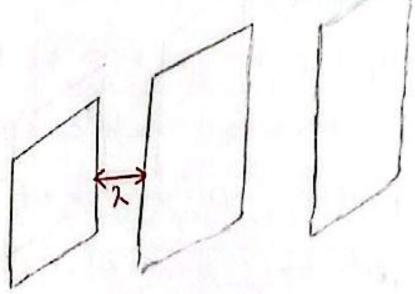
Spherical wave fronts

Those wavefrons which are of Spherical Shape.



rays Shows direction of propagation of light

plane wave fronts.



Those wavefronts which are of plane Shape.

RAYS
- → imaginary lines
- → Drawn "⊥" to the wave fronts.
- → Shows direction of propagation of light.

Wave front of Sun

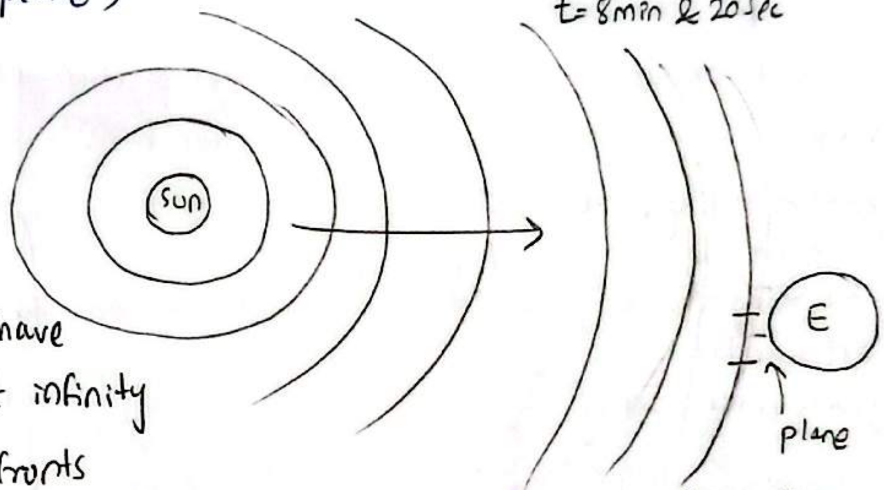
(Spherical)

appear on earth.
(plane)

Small
part of
Spherical wave
front

:Light covers 3lac distance
within a sec

t= 8min & 20sec



MCQ #(20) Near a Point Source we have ___Spherical___ wavefronts, while at infinity we always have ___plane___ wavefronts respectively.

(a) plane, Spherical

(b) $\boxed{\text{Spherical, plane}}$

(c) plane, plane

(d) Spherical, Spherical.

As wavefront reaches to Earth from Sun there Size increases to such an extent that they does not appear Spherical but rather plane.

For example: Earth is Spherical but when we see it appeared to us as it is plane this is because of its huge Size & we are looking at only a Small patch. which appeared plane to us.

· Tube Light:

↳ cylinderical wave fronts.

① plane, plane ...

② Spherical, Spherical -

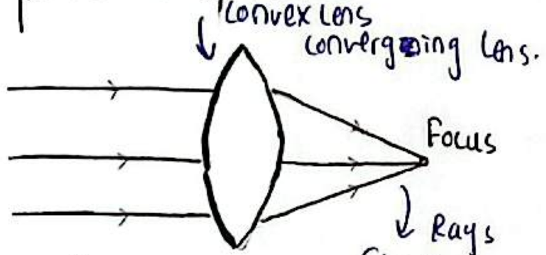
... example: Earth is Spherical but when we see it appeared to us as it is plane this is because of its huge size & we are looking at only a small patch. which appeared plane to us.

.Tube light:

↳ cylinderical wave fronts.

: Conversion of wave fronts:

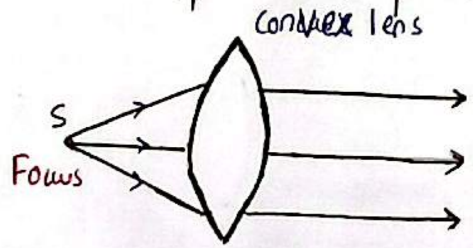
plane to spherical



↓ Rays shows plane wavefront

↓ Rays shows Spherical wavefront

Spherical to plane



if Source of light placed at focus of convex lens then all rays of light will get parallel.

Spherical wavefronts

↳ wavefront propagates with the Speed of light.

plane wave fronts

Huygen's PRINCIPLE (1678):

Location, $S = vt$
$\Delta x = c\,\Delta t$

used to determine.

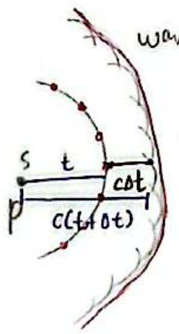
Shape of new wave front

Location or position of new wave front.

To obtain Shape of new wave front we join the tangents of all wavelets.
tangents = outerboundary

Acc. to Huygens assumption all these points will behave as Secondary Source of light.

Both wavefronts & wavelets travels with speed of light (c)



According to Huygens there are two types of Source of light.

primary Source of light → original Source of light → always produce wave fronts.

&

Secondary Source of light. → Points on wave fronts behave like Source of light. → produce wave lets → wavelets are Small Spherical curves.